Course Project: Endterm Rubric

Machine Learning Algorithms, 100 points

The Endterm develops your Midterm project on the same data and saved train/test split through tuned ensembles, unsupervised analysis and an MLP. Both partners upload Endterm_Surname_Name_Group.zip to Moodle by 23:59 on the day before the defense.

Text and image models are optional at this stage. Points go to decisions you can justify, fair comparisons and a careful look at errors. A high metric earns nothing by itself.

Criterion	Points	Scored for	What it checks
1. Continuity and fixes since the Midterm	10	team	The same data and split, explained fixes
2. Tuned ensembles	15	team	Forest and boosting tuned by training CV
3. Unsupervised analysis	10	team	Clustering or PCA/t-SNE, interpreted
4. Neural network against the best earlier model	15	team	MLP learning check and fair comparison
5. Fair comparison and error analysis across all models	15	team	Justified metrics and explained errors
6. Conclusions, limitations and plan for the Final	10	team	Conclusions and a text or image plan
7. Presentation within the time limit	10	team	A clear talk of 7 minutes
8. Individual answers at the defense	15	each student	Understanding, contributions and AI use
Total	**100**		

Scoring

Each criterion gets a level from 0 to 4 with the fixed points below. Criteria 1-7 are scored once for the team. Each student's grade is the team points out of 85 plus their own criterion 8 points out of 15.

The defense lasts 12 minutes per team, with 7 minutes for the presentation and 5 minutes for questions. Both partners speak, and either partner may be asked about any part of the project.

AI tools are allowed. Say which tools you used and for what, and be ready to explain every part of your work yourself.

Criterion weight	4 Excellent	3 Good	2 Adequate	1 Weak	0 Missing
20 points	20	15	10	5	0
15 points	15	11	8	4	0
10 points	10	8	5	3	0

A complete submission contains

Both partners upload the same complete archive. Include everything shown at the defense.

Element	Criterion
Code with outputs, change log, data and saved split	1-5
Tuning, unsupervised and MLP checks, results and errors	2-5
Conclusions, limitations and Final source plan	6
Slides for the 7-minute talk	7
Who did what and AI tools used, or a statement of no use	8
Run instructions	1-5

Levels for each criterion

Criterion	4 Excellent	3 Good	2 Adequate	1 Weak	0 Missing
1. Continuity and fixes since the Midterm 10 points, team	**10 points** Uses the same data and saved train/test split as the Midterm. A short change log links feedback and observed weaknesses to specific fixes, with before/after evidence. Restart and run all reproduces the numbers reported in the slides and notebook, apart from small hardware differences. Remaining issues are named honestly.	**8 points** Keeps the same data and split. Most fixes are explained and checked, but evidence for one change is brief.	**5 points** Shows some fixes, but the link to Midterm feedback or before/after evidence is thin. Split continuity is only partly documented.	**3 points** Shows a few edits with little explanation. The saved split or the effect of the fixes is unclear.	**0 points** Provides no traceable continuation of the Midterm work.
2. Tuned ensembles 15 points, team	**15 points** Tunes both random forest and boosting with cross-validation on training data only. Preprocessing is fitted within training folds. Explains the search space and selected settings. Compares both with the Midterm best model on the saved split using the same metric and reports CV mean and spread.	**11 points** Tunes both ensembles on training folds and compares them fairly with the Midterm best model. Search choices or CV variability receive little explanation.	**8 points** Both ensembles run, but neither or only one has a documented CV search, or the Midterm comparison is incomplete.	**4 points** Only one ensemble runs, or tuning and evaluation use the test set, so the comparison is unreliable.	**0 points** Provides no usable random forest or boosting results.
3. Unsupervised analysis 10 points, team	**10 points** Uses clustering or PCA/t-SNE on the project objects. Justifies scaling and key settings. Interprets clusters, components or a map using actual object features and examples. Checks relevant stability or explained variance and describes the limits of the structure without treating a map as proof of real groups. Connects the structure to the Midterm models: tests whether PCA components or cluster labels, fitted on training folds, change the best model's score, or shows how the groups explain where that model fails.	**8 points** Method, scaling and interpretation are sound, and examples support the claims. The stability check or limitations are brief, or the connection to the Midterm models is missing or incomplete.	**5 points** Produces clusters or a projection, but scaling choices and the meaning for the objects are only partly explained.	**3 points** Shows a plot or labels with little interpretation, or claims that visible groups prove categories without evidence.	**0 points** Provides no usable clustering or dimensionality reduction analysis.

4. Neural network against the best earlier model 15 points, team	**15 points** Trains an MLP with train-only preprocessing and explains its architecture and settings. Checks training and validation learning curves or a validation plateau before comparison. Compares with the best earlier model, including the Midterm models and tuned ensembles, on the same saved split and metric. Explains overfitting, convergence and any lack of improvement.	**11 points** Checks the MLP learning progress and compares fairly with the best earlier model. Architecture choices or the interpretation of the curves are brief.	**8 points** The MLP and comparison run, but learning progress is not checked, or the earlier reference is not clearly the best model.	**4 points** Shows a partial MLP run, or uses different test rows or test-guided training, leaving the comparison unreliable.	**0 points** Provides no usable MLP result.
5. Fair comparison and error analysis across all models 15 points, team	**15 points** Justifies the main metric and compares the baseline, Midterm models, ensembles and MLP in one table with CV mean and spread and test scores on the saved split. Keeps model selection on training data. Examines residuals or a confusion matrix and real errors across models. Explains trade-offs and where each model fails. The conclusion states what changed since the Midterm and why.	**11 points** The comparison is complete and fair, with real error examples. Explanations of error patterns or model trade-offs are brief.	**8 points** Compares most models, but CV variability or error examples are missing. Conclusions mainly repeat metric values.	**4 points** Shows scattered scores with incompatible splits or metrics, or interprets errors incorrectly.	**0 points** Provides no traceable comparison or error analysis.
6. Conclusions, limitations and plan for the Final 10 points, team	**10 points** Answers the project question and ties limitations to the data and observed errors. Names the text or image source to add for the Final, explains lawful access and how it links to the same objects, and says why it may address a current weakness. Gives a feasible next step and explains how its value will be tested.	**8 points** Conclusions and limitations follow the results. Names a suitable text or image source, but linkage or the planned comparison needs detail.	**5 points** Gives conclusions and a source idea, but the limits are generic or the reason for adding that source is weak.	**3 points** Offers a short conclusion and a vague plan with little connection to the results.	**0 points** Provides no conclusions or Final plan.
7. Presentation within the time limit 10 points, team	**10 points** Uses the 7 minutes for a clear talk that moves from the question to the next steps. Slides are readable, figures are labelled, and both partners speak.	**8 points** Clear overall but uneven: code on the slides, unlabelled figures or timing a little off.	**5 points** Hard to follow, or the main result is missing.	**3 points** Far over time, with no clear line through the talk.	**0 points** Gives no presentation.

8. Individual answers at the defense 15 points, each student	**15 points** Explains any part of the project, including the partner's work, and answers questions about tuning, unsupervised structure, MLP learning and errors correctly. Uses the submitted 'who did what' section to explain their contribution. Names AI tools and their uses, or states that none were used, and can explain the resulting work independently.	**11 points** Explains most of the whole project and their documented contribution. AI use is disclosed clearly. A few method answers need prompting.	**8 points** Explains their own part and some methods, but has gaps elsewhere. Contributions or AI use are described only partly.	**4 points** Answers a few basic questions and gives little account of their contribution or AI-assisted work.	**0 points** Cannot explain the submitted work or their contribution.